1. Observation (Audit)

> *“Measure first, optimize next. Systems that are not observable cannot be optimized or scaled.”*

This document is to observe and recollect what are all the existing workloads running, the operational costs, resources consumption, environments, nature of the workloads, workloads categorization, identify stale resources - focuses on establishing **complete visibility** into the organization’s infrastructure across AWS and on-prem environments.

With all that, will make decisions on

- Identify **all running workloads**
- Understand **resource utilization patterns**
- Analyze **cost distribution**
- Classify workloads based on **function and behavior**
- Detect **inefficiencies, unused, and stale resources**

This serves as the **baseline for optimization and modernization decisions**.

Infrastructure - AWS

1. Resources Summary

Summary of your resources across all Regions in our organization's account.

Reference: Global Resources List, Global Resources Search

Overall Footprint:

Compute	Storage	Networking	Security
229	328	1148	418

Deepdive:

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- Enabled regions: 29 regions
- Instances: 226 in 4 regions
- VPCs: 35 in 28 regions
- Subnets: 104 in 28 regions
- Security groups: 382 in 28 regions
- Volumes: 260 in 5 regions
- Auto scaling groups: 3 in 1 regions
- Route tables: 44 in 28 regions
- VPC endpoints: 3 in 1 regions
- NAT gateways: 2 in 1 regions
- Egress only internet gateways: 0 in 0 regions
- Internet gateways: 34 in 28 regions
- DHCP option sets: 28 in 28 regions
- Elastic IPs: 234 in 3 regions
- Endpoint services: 0 in 0 regions
- Managed prefix lists: 338 in 28 regions
- Network ACLs: 36 in 28 regions
- Network interfaces: 336 in 4 regions
- VPC peering connections: 1 in 1 regions
- Capacity Reservations: 0 in 0 regions
- S3 buckets: 62 in 2 regions
- RDS clusters: 2 in 1 regions
- RDS DB instances: 4 in 1 regions
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> **Signs of:**
> **- Heavy networking footprint → typical for telco workloads**
> **- Heavy hit rate costs**
> **- Potential over-provisioning or fragmentation (identified using Cost Optimization Hub Reports)**

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2. Cost and usage Summary

Ref: Cost Explorer

Total cost - $143,234.39
Last month cost - $31,144.10
Average monthly cost - $23,872.40
Active Services count - 35

Monthly cost spike percentage:

Last 3 Months Average (Jan–Mar 2026)==: $24,111.12 / month==

> “The current quarterly average spend indicating a **42.7% increase in cloud expenditure** from last year three month surge”

Last month cost - $31,144.10 (Bills Estimate extracted at 02/04/2026)

Service	Service total	January 2026	February 2026	March 2026*
Total costs	$143,234.39	$23,277.40	$28,926.25	$31,144.10
EC2-Instances	$49,960.95	$8,266.75	$13,506.56	$14,865.98
Relational Database Service	$24,279.26	$3,777.58	$4,396.28	$5,046.20
Tax	$19,586.30	$3,183.11	$3,955.39	$4,258.93
S3	$13,588.51	$2,261.67	$2,374.32	$2,221.65
Savings Plans for Compute usage	$11,857.36	$2,051.72	-	-
EC2-Other	$7,342.32	$1,156.12	$1,269.54	$1,308.29
VPC	$5,926.67	$986.01	$961.19	$1,137.63
Elastic Load Balancing	$2,218.16	$393.91	$351.50	$385.17
CloudWatch	$1,676.59	$223.66	$297.42	$539.72
Lambda	$1,630.46	$109.88	$809.20	$334.72
API Gateway	$1,228.51	$166.12	$331.84	$274.79
Route 53	$1,221.04	$206.67	$189.63	$206.40
DynamoDB	$747.23	$121.58	$130.89	$148.69
CloudFront	$687.27	$118.92	$121.03	$110.84
Elastic Container Service for Kubernetes	$664.59	$148.80	$134.40	$146.60
Direct Connect	$293.66	$49.70	$44.99	$51.88
EC2 Container Registry (ECR)	$138.58	$25.42	$30.87	$41.63
ElastiCache	$113.53	$14.88	$13.44	$41.05
Registrar	$23.00	$8.00	-	$15.00
Secrets Manager	$16.67	$2.80	$2.72	$2.76
Polly	$12.32	$0.29	$0.15	$0.30
Key Management Service	$12.04	$2.01	$2.01	$1.98
Systems Manager	$4.42	$1.43	$1.42	$1.55
Elastic File System	$4.11	$0.29	$1.35	$1.99

Service	Service total	January 2026	February 2026	March 2026*
Lightsail	$0.56	$0.10	$0.10	$0.09
SQS	$0.25	$0.00	$0.00	$0.25
SNS	$0.01	$0.00	$0.00	$0.00
Backup	$0.00	$0.00	$0.00	$0.00
CloudShell	$0.00	-	$0.00	$0.00
Step Functions	$0.00	-	$0.00	$0.00
CloudTrail	$0.00	$0.00	$0.00	$0.00
Glue	$0.00	$0.00	$0.00	$0.00
Location Service	$0.00	$0.00	-	-
CloudWatch Events	$0.00	$0.00	$0.00	$0.00
X-Ray	$0.00	-	$0.00	-

> "The current quarterly average spend indicating a **42.7% increase in cloud expenditure** from last year three month surge" Steady surge.

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3. EBS and Data Summary:

Ref: Search AWS Resources

- General purpose buckets (62)
- EBS Volumes
- SnapShots (547)

> Chunk of essential and idle data in the mix.

Infrastructure - On-prem

Self Managed by administrators.

App Workloads:

Status:

Production is up and running on AWS with a mix of EC2 (monolith), EKS(microservices) and other managed services supporting in the backend, having RDS, Aurora DBs as Databases. Plus, there are static workloads for big ticket customers which should be touched or maybe if needed as per management or customer's request. (Customers: GuideHost, Casagrande, Gebbs)

> **Currently in the process of minimally migrating the traffic from legacy systems to the latest ConnectHub - a Unified backend service.**

Applications:

On EC2:

1. **PPX** - Legacy System - Monolith running in EC2 Instances - **10+ Instances**
2. **Call Desk** - **20 to 22+ EC2 Instances**
    1. Call Spot
    2. Marketoff
    3. JC
    4. Guidehouse

On EKS:

1. **ConnectHub** - EKS 1cluster (6Nodes), EC2 - 9VMs (Static stateful workloads) + 5 - 50 EC2 Instances at scale.
2. Sandbox Cluster for Testing - 1 EKS Cluster.

**Each systems has been setup on each dedicated EC2 instances of each of the customers.

Currently in process of migrating specific customers traffic at a minimal scale from Legacy PPX or Call Desk systems to the Modern and unified System - Connecthub.

Out of all that, there are SIP broadband workloads running on hybrid (on-prem and cloud) in the mix.

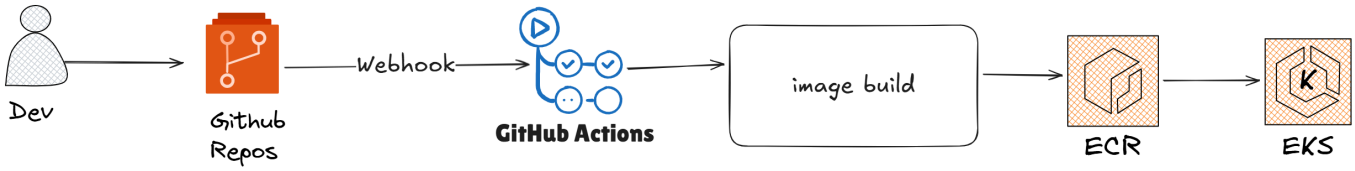
Tech Stack:

1. LAMP
2. Python - Flask, Django
3. DBMS - Adminer, Phpmyadmin
4. WebSockets (RAW)
5. and more

Services (Telecom Systems):

1. Synapse
2. FlexiSIP
3. openSIPs
4. Routr
5. Kamailio
6. Rocketchat
7. Asterisk
8. Freeswitch and more

Developer Workflow:



Verdict:

Infra:
Current trends indicate a **~42% increase in cloud spend**, primarily driven by:

- Compute-heavy workloads (EC2)
- Persistent database usage (RDS)
- Network/data transfer costs (VPC, bandwidth-heavy RTP traffic)

DevWorkflow:

- Straightforward CI/CD
- No SecOps
- Resource Segregation clutter
- Manual Implementation (No IaC)
- Prone to resources going stale and being residue
- Resource spec mismatch
- Lack of Audit